2020A F=ma Exam: Problem 7

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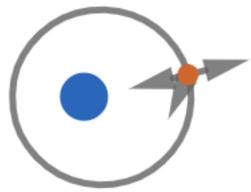

Recall the energy of an elliptical orbit is given by

$$E = -\frac{GMm}{2a}$$

The energy delivered to the trash by the astronaut's throw is negligible compared to its initial energy since

$$\Delta v \ll \sqrt{GM/a}$$

where $\sqrt{GM/a}$ is on the order of the orbital velocity. Since E doesn't change, a doesn't change and by Kepler's third law $T^2 \propto a^3$, all pieces of trash have the same period as the space station and return after a single orbit. Thus, the answer is $\boxed{\text{E}}$.